ARDHRA.K

AI ML ENGINEER

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SUMMARY

Machine Learning and AI Developer with strong foundation in designing, developing, and deploying ML/DL models across Computer Vision, NLP, and predictive analytics. Skilled in building CNNs, working with Transformers and LLMs, and optimizing model performance through hyperparameter tuning and evaluation. Proficient in Python, TensorFlow, PyTorch, scikit-learn, and OpenCV. Adept at data preprocessing, feature engineering, and delivering scalable, high-impact solutions using modern ML frameworks and tools.Experienced in end-to-end machine learning pipelines, from data collection and preprocessing to model deployment and performance monitoring in real-world applications.

Education

Bachelor of Computer Applications (BCA),SNDP College, Perinthalmanna 2025
CGPA: 7.35 / 10.0
Class 12th , Kerala Board of Public Examination 2022
Percentage: 79%
Class 10th ,Central Board of Secondary Education 2020
Percentage: 78.6%

skills

Python, SQL, MongoDB, HTML5, CSS3, Pandas, Flask, TensorFlow, Machine Learning, Artificial Intelligence, PyTorch, Power BI, Tableau, Data Science, Version Control (Git), Adaptability, Teamwork, Problem-Solving, OpenCV, NLP, Transformers, LLM, Computer Vision, Keras, Git, Hugging Face, RAG, LangChain
Programming Skills: Python
Database Skills: SQL, MongoDB
Technologies: NumPy, Pandas, Flask, Machine Learning, Artificial Intelligence, PyTorch, TensorFlow, OpenCV, NLP, Transformers, LLM, Computer Vision, Keras, Hugging Face, RAG, LangChain
Tools & Concepts: Power BI, Tableau, Data Science, Version Control (Git)
Soft Skills: Adaptability, Teamwork, Problem-Solving

PROJECT

Handwritten Digit Recognition

- Preprocessed 70,000 MNIST handwritten digit images using normalization, reshaping, and one-hot encoding
- Designed and implemented a CNN architecture with convolutional, pooling, dropout, and fully connected layers using TensorFlow/Keras
- Optimized model performance through hyperparameter tuning (epochs, batch size, filters, kernel size)
- Achieved ~98–99% test accuracy with evaluation using precision, recall, F1-score, and confusion matrix
- Built a real-time digit classification system for user-drawn digits
- Conducted error analysis on misclassified samples to improve model robustness

Liver Patient Multi Model Prediction

- Processed a medical dataset of 582 records with 11 features, handling missing values in Albumin_and_Globulin_Ratio via mean imputation
- Performed data cleaning and EDA: removed duplicates, label encoding, and analyzed feature correlations and class imbalance
- Selected and compared multiple tree-based classifiers (Random Forest, Gradient Boosting, XGBoost)
- Applied hyperparameter tuning (Random Search) on Gradient Boosting
- Achieved results: Accuracy: 0.71, Precision: 0.71, Recall: 1.00, F1-score: 0.83
- Generated confusion matrix and classification reports for evaluation